Supplementary Material S1: WildMech codebook (taxonomy v1.0)

Each opening request receives one primary code (L1), optionally one L2 subcategory, and three facets. UNK requests are excluded from task shares.

Code	Category	Design phase	Definition	L2 subcategories
STD	Standards, codes and regulations	Requirements and standards	Find, interpret or comply with a standard, code or regulation (ISO, ASME, MIL, SAE, EN, DIN, API, FDA, internal design rules).	find_standard, interpret_clause, compliance_check, internal_design_rules
MEC	Concept and mechanism design	Concept and architecture	Generate or compare concepts, mechanisms, architectures or layouts for a function.	concept_generation, mechanism_synthesis, architecture_tradeoff, reverse_engineering
GEN	Generate a design or CAD model	Concept and architecture	Ask the AI to produce a design artifact for a described object: a 3D/CAD model, an assembly, a concept image or render, or a modification of a generated model. Includes 'design me X' requests for a physical product.	text_to_cad, image_or_drawing_to_cad, concept_image, iterate_generated_model, whole_product_design
CAL	Engineering calculations and analysis	Embodiment and sizing	Hand calculation or analytical estimate: strength, fatigue, thermal, fluids, dynamics, vibration, kinematics, buoyancy, unit conversion.	structural, fatigue_fracture, thermal, fluids, dynamics_vibration, kinematics, units_conversion, other
CMP	Component selection and sizing	Embodiment and sizing	Choose or size an off-the-shelf component: bearing, motor, gearbox, spring, actuator, sensor, heat sink, valve, linear guide.	bearings_guides, motors_actuators, springs, thermal_components, sensors, fluid_components, other
MAT	Materials, coatings and treatments	Embodiment and sizing	Select a material, alloy, coating, finish or heat treatment; corrosion, compatibility, properties.	material_selection, properties_lookup, coating_finish, heat_treatment, corrosion_compatibility
JNT	Joints, fasteners and sealing	Embodiment and sizing	Bolted, welded, bonded, threaded and snap joints; preload and torque; O-rings, gaskets and seals.	bolted_threaded, welded_bonded, snap_press_fit, seals_orings
SIM	Simulation setup and interpretation	Embodiment and sizing	Set up, run or interpret FEA, CFD, thermal or multibody simulation.	fea, cfd, thermal_sim, multibody, results_interpretation
GDT	Tolerancing, fits and GD&T	Detail design and documentation	Tolerances, fits, GD&T callouts, tolerance stack-ups, process capability.	fits, gdt_callouts, stackup, process_tolerances
CAD	CAD, PDM and PLM how-to	Detail design and documentation	How to do something in a CAD, PDM or PLM tool; modeling technique; automation and macros.	modeling_technique, drawings_tool, pdm_plm, automation_api
SRCH	Find internal parts and knowledge	Detail design and documentation	Find an existing part, assembly, drawing, document or past decision inside the company's own data.	part_reuse, document_search, past_decision, internal_standard
DOC	Documentation, drawings and BOM	Detail design and documentation	Write or check drawings, BOMs, specs, reports, ECOs, work instructions, emails.	drawing_check, bom, report_spec, eco_change, communication

Code	Category	Design phase	Definition	L2 subcategories
DFM	Manufacturing processes and DFM/DFA	Manufacturing and sourcing	Process choice and limits (machining, sheet metal, molding, casting, AM, welding), DFM/DFA rules, process conversion, cost drivers.	machining, sheet_metal, molding_casting, additive, process_conversion, assembly_dfa
SUP	Sourcing, suppliers and cost	Manufacturing and sourcing	Find a supplier or catalog item, alternatives, availability, lead time, cost estimate.	supplier_search, alternatives, cost_estimate
TST	Testing, validation and qualification	Verification and in-service	Plan or interpret tests, qualification, inspection and quality checks.	test_planning, test_standards, inspection_quality, data_analysis
FAIL	Troubleshooting and failure analysis	Verification and in-service	Diagnose a failure, defect, mismatch or test-versus-model discrepancy; root cause.	field_failure, manufacturing_defect, model_test_mismatch, assembly_issue
EDU	Learning and coursework	Learning and coursework	Explain an engineering or physics concept with no design task attached, or solve a textbook, exam or homework-style problem.	concept_explanation, textbook_problem, literature_pointer, study_planning
OOS	Outside mechanical design	Outside mechanical design	Electronics, software, architecture and buildings, business writing, translation, general knowledge, non-engineering homework, and questions about the assistant itself.	electronics, software_code, buildings_architecture, writing_business, general, about_the_assistant
UNK	Not classifiable on its own	Outside mechanical design	Greeting, bare continuation of an earlier exchange, reply to the assistant's clarifying question inside an agentic task, or complaint with no task. Excluded from task shares and reported separately.	continuation, greeting, agent_flow_reply, complaint

Facets

Complexity
C1: Single fact or lookup
C2: One rule or one calculation
C3: Multi-step reasoning or trade-off between options
C4: Open-ended design judgment that needs context the query does not hold

Context needed: none; the engineer's geometry (attached or open model, drawing, photo); the company's internal data; both.
Verifiable answer: yes if a checkable numeric or closed-form answer exists.

Decision rules for frequent confusions

Create, model, design or draw a physical object, or change a generated model: GEN. Ideas, mechanisms or an approach in words: MEC. Choose or size a component type from requirements: CMP. Locate a specific part number, vendor, price or catalog CAD file: SUP. Practitioner calculation: CAL. Textbook, exam or homework problem, or a concept explanation with no design task: EDU. Questions about the assistant itself, and topics outside mechanical design: OOS. Greetings, bare continuations, replies to the assistant's clarifying questions, complaints with no task: UNK.